

Advanced Dynamics of Mechanical Systems

Assignment 1 – Part A
Experimental Modal Analysis of a Cantilever Beam

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1 Objective and reference structure

This section, designated as Part A, presents a comprehensive modal analysis of a slender aluminium cantilever beam. The study is structured around three primary objectives:

1. **Analytical Modeling:** The natural frequencies and corresponding mode shapes are derived analytically, leveraging the standing-wave solution based on the Euler–Bernoulli beam theory in bending vibration.
2. **FRF Synthesis:** The forced dynamic response of the system is evaluated by computing the Frequency Response Functions (FRFs) for several chosen input and output coordinates.
3. **Modal Parameter Identification:** Treating these numerically generated FRFs as surrogate experimental data, the system’s modal parameters such as natural frequencies (ω_n), damping ratios (ζ_n), and mode shapes are recovered employing an FRF-based single-mode curve-fitting method ($n = 1$).

The structural properties and geometric parameters of the reference beam are detailed in Table 2.

Table 1: Geometry and material properties.

Parameter	Symbol	Unit	Value
Length	L	mm	1200
Thickness	h	mm	8
Width	b	mm	40
Density	ρ	kg m^{-3}	2700
Young’s modulus	E	GPa	68

Derived quantities: mass per unit length $m = \rho b h$, second moment of area $J = \frac{1}{12} b h^3$ and bending stiffness EJ .

Table 2: Derived quantities.

Parameter	Symbol	Unit	Value
Mass per unit length	m	kg m^{-1}	0.864
Second moment of area	J	m^4	1.7067×10^{-9}
Bending stiffness	EJ	Nm^2	116.0533

2 Natural frequencies and mode shapes

2.1 Procedure

The free transverse vibration of a slender beam is governed by the Euler-Bernoulli equation:

$$EJ \frac{\partial^4 w}{\partial x^4} + m \frac{\partial^2 w}{\partial t^2} = 0$$

A standing-wave ansatz, separating the spatial and temporal variables as $w(x, t) = \phi(x)g(t)$, yields a harmonic time law and the spatial eigenvalue equation:

$$\phi''''(x) - \gamma^4 \phi(x) = 0, \quad \gamma^4 = \frac{\omega^2 m}{EJ}$$

The general solution to this spatial differential equation is a combination of trigonometric and hyperbolic functions:

$$\phi(x) = A \cos(\gamma x) + B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)$$

To find the mode shapes of our system, we must apply the boundary conditions of a cantilever beam (clamped at $x = 0$ and free at $x = L$). Imposing these conditions on $\phi(x)$ and its derivatives generates a homogeneous linear system of equations, which can be expressed in matrix form as $[H(\omega)]\mathbf{z} = \mathbf{0}$, where $\mathbf{z} = \{A, B, C, D\}^\top$ is the vector of unknown coefficients.

At this point, the system always admits the *trivial solution* ($\mathbf{z} = \mathbf{0}$), which physically represents the beam perfectly standing still. However, since we are interested in the dynamic vibrating behavior of the structure, we must look for non-trivial solutions.

A non-zero vector $\mathbf{z}$ exists only if the matrix $[H(\omega)]$ is singular. Therefore, to find the vibrating solutions, we force the determinant of the matrix to be zero:

$$\det[H(\omega)] = 0$$

Solving this characteristic equation numerically yields a discrete set of roots, which correspond to the system's natural frequencies ω_n . Finally, substituting each natural frequency back into the rank-deficient system, such that $[H(\omega_n)]\mathbf{z}^{(n)} = \mathbf{0}$, allows us to determine the relative values of the coefficients and compute the corresponding mode shape $\phi_n(x)$.

Numerical implementation. To numerically locate the natural frequencies, the determinant of the boundary condition matrix, $\det[H(\omega)]$, is evaluated across a defined frequency grid. The system's natural frequencies correspond to the points where the absolute value of the determinant reaches its local minima (i.e., approaching zero), as illustrated in Fig. 1.

Because the evaluated matrix $[H(\omega_n)]$ is rank-deficient, the system admits infinite proportional solutions. To resolve this, one of the four coefficients is arbitrarily fixed, which reduces the problem to a fully determined 3×3 linear system.

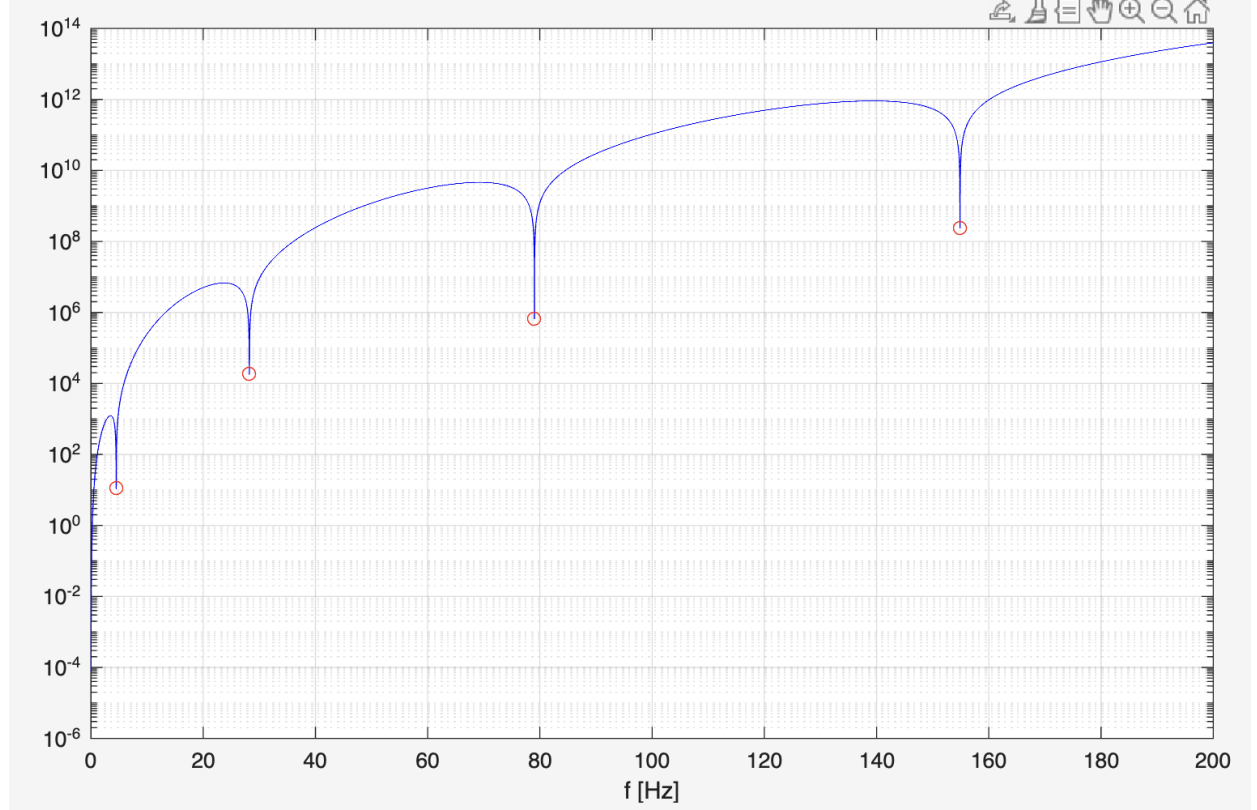


Figure 1: Absolute value of the determinant, $|\det[H(\omega)]|$, versus frequency on a semilogarithmic scale. Red markers indicate the first four natural frequencies. Although analytically the determinant evaluates to exactly zero at these frequencies, the use of a discrete frequency vector in the MATLAB implementation requires us to approximate these roots by identifying the local minima.

Table 3: Identified natural frequencies.

	Mode 1	Mode 2	Mode 3	Mode 4
Identified Frequency [Hz]	4.50	28.22	79.03	154.86

2.2 Obtained mode shapes

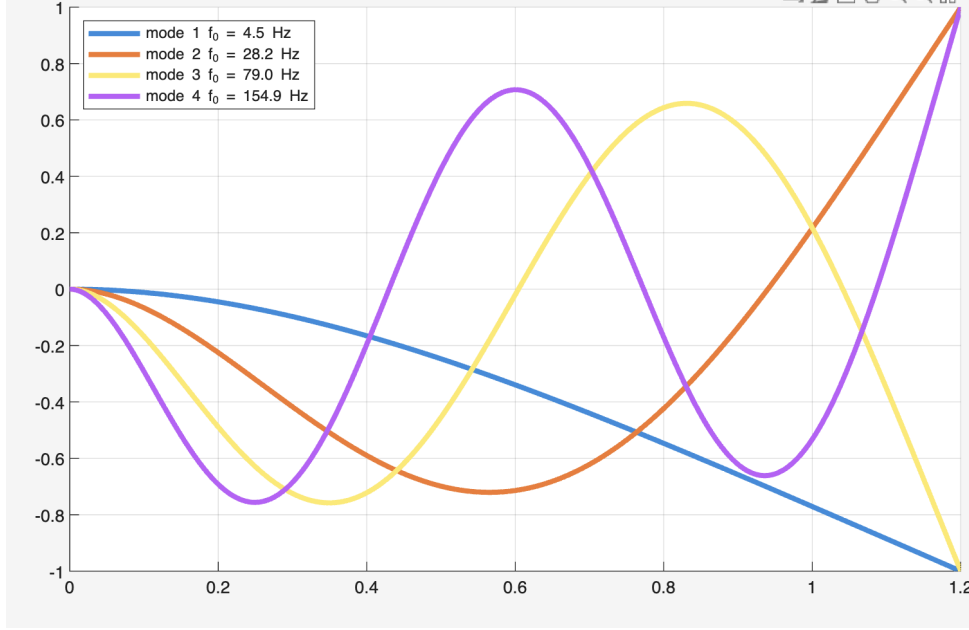


Figure 2: First four normalised bending mode shapes and natural frequencies.

Comments.

- *Mode shape computation.* The mode shapes were computed by substituting the identified natural frequencies back into the analytical standing-wave function.
- *Boundary conditions.* Looking at the plotted mode shapes, we can verify that the physical constraints at the fixed end are correctly respected. In fact, for all the evaluated modes, both the displacement and the slope (the first derivative) are strictly equal to zero at the clamp ($x = 0$).
- *Mode shape evolution.* By observing the mode shapes along the entire span of the beam, we can notice a typical dynamic behavior: as the mode number and the associated natural frequency increase, the number of lobes increases as well. Consequently, the higher modes show more zero-crossing points (nodes) along the beam, perfectly matching the expected analytical results.
- *Normalization.* Finally, it is important to point out that the plotted mode shapes have been normalized to be easily compared on the same plot. Specifically, each mode shape vector was divided by its absolute maximum value, ensuring that the maximum amplitude for every mode is always scaled to exactly 1 or -1.

3 Frequency Response Functions

3.1 Procedure

To compute the forced dynamic response of the cantilever beam, we rely on the principle of modal superposition. Thanks to the orthogonality property of the mode shapes, the complex continuous system can be decoupled and treated as a linear combination of independent single-degree-of-freedom oscillators. First, the modal mass m_n for each mode was computed by numerically integrating the mode shapes along the beam using the trapezoidal rule (via MATLAB's `trapz` function):

$$m_n = \int_0^L m \phi_n^2(x) dx$$

To properly model the dynamic behavior, a structural modal damping ratio of $\zeta_n = 1\%$ was assigned to all the evaluated modes. The Frequency Response Function (FRF), which represents the receptance between an input force applied at a coordinate x_k and the corresponding displacement measured at x_j , is calculated by summing the individual contributions of each mode:

$$G_{jk}(\Omega) = \sum_{n=1}^N \frac{\phi_n(x_j) \phi_n(x_k)}{m_n(\omega_n^2 - \Omega^2 + i 2\zeta_n \omega_n \Omega)} \quad (1)$$

It is worth noting that the ratio $\phi_n(x_j)\phi_n(x_k)/m_n$ is intrinsically independent of the normalization chosen for the mode shapes. Finally, to evaluate the sensitivity of the identification method, two different measurement configurations are analyzed. The first setup employs an input force located at 0.2 m and three output positions situated in roughly antinodal regions ($x_j = \{0.4, 0.7, 1.2\}$ m). This choice ensures a strong modal participation and well-defined resonance peaks in the FRFs. Conversely, in the second setup, we intentionally challenge the identification process to observe potential estimation errors. To do so, the output positions are moved near the structural nodes of the mode shapes ($x_j = \{0.2, 0.6, 1.08\}$ m); this forced condition allows us to evaluate the method's limitations when a specific mode has a near-zero amplitude and its contribution to the overall response practically vanishes.

3.2 Results and comments

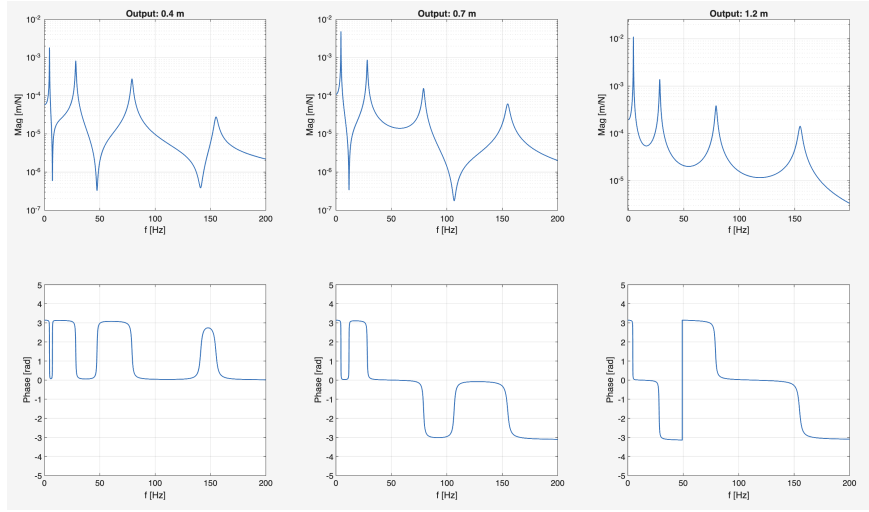


Figure 3: Reference FRFs: magnitude (top) and phase (bottom) for the force at $x_k = 0.2$ m and outputs $x_j = 0.4, 0.7, 1.2$ m.

Comments.

- *Resonances (poles)*. As the chosen output coordinates ($x_j = \{0.4, 0.7, 1.2\}$ m) are deliberately located in roughly antinodal regions—where the modal displacements for the first four modes are substantial—the modal participation is highly effective. Consequently, all the evaluated FRFs exhibit sharp and well-defined resonance peaks. These peaks occur exactly at the system’s natural frequencies, visually confirming that the poles of the transfer function are intrinsic, global structural properties, entirely independent of the spatial locations of the input and output nodes.
- *Antiresonances (zeros)*. The deep notches visible between some consecutive peaks represent the antiresonances (zeros) of the system, arising from the destructive interference between adjacent modes.
 - At $x_j = 0.4$ m, the condition for destructive interference is successfully met between all the first four modes. This creates three distinct antiresonances (deep notches) separating every single resonance peak.
 - At $x_j = 0.7$ m, the antiresonance between Mode 2 and Mode 3 (around 50-60 Hz) completely disappears. This happens because the mode shape ϕ_3 exhibits a sign inversion compared to ϕ_2 at this specific spatial coordinate, flipping the sign of the modal constant and preventing cancellation.
 - Finally, at the free tip ($x_j = 1.2$ m), the modal constants alternate in sign for every consecutive mode. Therefore, the equal-sign condition is never met, resulting in an FRF completely devoid of intermediate antiresonances.
- *Phase behavior*. The phase diagrams perfectly mirror the pole-zero sequence of the magnitude plots. Each lightly damped resonance (complex pole pair) induces a steep phase lag of π radians, whereas each antiresonance (zero pair) compensates with a phase lead of π .
- *Phase behavior*. It is worth noting that all phase plots originate at a value of π rad (180°). This is a direct consequence of the chosen reference system: the applied input force is directed downwards, while the measured output displacement is defined upwards. This physical sign inversion results in a negative static receptance, which corresponds to an initial phase of π . Moving along the frequency axis, the phase diagrams perfectly mirror the pole-zero sequence of the magnitude plots. Each lightly damped resonance (complex pole pair) induces a steep phase lag of π radians, whereas each antiresonance (zero pair) compensates with a phase lead of π .

3.3 Results and comments

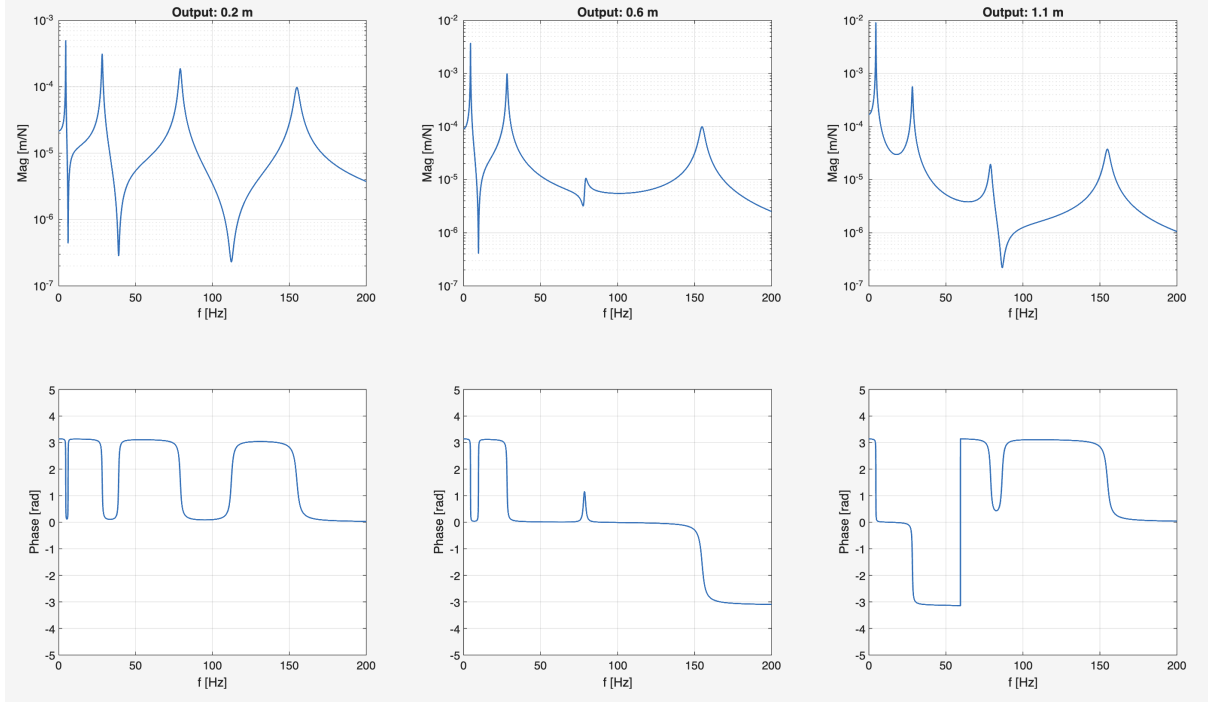


Figure 4: Reference FRFs: magnitude (top) and phase (bottom) for the force at $x_k = 0.2$ m and outputs $x_j = 0.2, 0.6, 1.1$ m.

Comments.

- *Resonances and Nodal Proximity.* In this alternative configuration, the output coordinates ($x_j = \{0.2, 0.6, 1.1\}$ m) were intentionally selected to challenge the identification algorithm by placing the sensors closer to the structural nodes of the mode shapes. This choice drastically affects the amplitude of the resonance peaks. For instance, in the measurement at $x_j = 0.6$ m, the third mode is almost completely hidden; this occurs because 0.6 m coincides almost perfectly with a zero-crossing point (node) of the third mode shape $\phi_3(x)$.
- *Antiresonances and Co-located FRF.* The measurement at $x_j = 0.2$ m is of particular theoretical interest because it represents a *co-located* (or drive-point) FRF, since the output coordinate exactly matches the input location ($x_k = x_j = 0.2$ m). In a co-located setup, the modal constant $A_n \propto \phi_n(x_k)\phi_n(x_j)$ becomes proportional to the square of the mode shape, $\phi_n(0.2)^2$, which is inherently strictly positive for all modes. Because the modal constants of all adjacent modes share the exact same sign, complete destructive interference is mathematically guaranteed to occur between *every* pair of consecutive peaks. The plot confirms this property, displaying a deep antiresonance notch between peaks 1-2, 2-3, and 3-4.
- *Phase behavior.* As previously established, all phase plots originate at π due to the sign inversion between the downward input force and the upward output displacement convention. The phase profiles perfectly map the pole-zero dynamics. For the co-located FRF (0.2 m), the strict and systematic alternation of poles (inducing a π phase lag) and zeros (inducing a π phase lead) generates a remarkably regular, repeating sequence of drops and recoveries. Conversely, in the 0.6 m output, the near-zero amplitude of the third mode and its immediately adjacent antiresonance almost cancel each other out dynamically; this results in a tiny, anomalous "spike" in the phase plot, completely lacking the standard full π rad transition.

4 Modal parameter identification

4.1 Method

For a lightly damped structure, in a band around ω_n the FRF is dominated by that mode; the remaining modes contribute only slowly varying tails, captured by two residual terms,

$$H_{jk}^{\text{fit}}(\Omega) \approx \frac{A_{jk}^{(n)}}{\omega_n^2 - \Omega^2 + i 2\zeta_n \omega_n \Omega} + \frac{R_{jk}^L}{\Omega^2} + R_{jk}^H, \quad (2)$$

with $A_{jk}^{(n)} = \phi_n(x_j)\phi_n(x_k)/m_n$ the modal constant. The term R^L/Ω^2 is the inertial (mass-like) tail of the modes *below* the band (at Ω above their resonance each behaves as $\sim -A/\Omega^2$), and R^H is the quasi-static (stiffness-like) tail of the modes *above* the band (at Ω below their resonance each behaves as $\sim A/\omega^2 = \text{const}$). The residuals are not modal parameters: they absorb the out-of-band tails so the resonant term fits the peak alone. *If they were omitted*, the resonant term would be forced to absorb the offset and slope of the neighbouring modes, biasing the identified $A_{jk}^{(n)}$ (and slightly ω_n, ζ_n).

4.2 Least-squares fit and initial guesses

The parameters minimise the complex least-squares functional

$$\varepsilon = \sum_j \sum_i \left\{ \text{Re}[H^{\text{exp}} - H^{\text{fit}}]^2 + \text{Im}[H^{\text{exp}} - H^{\text{fit}}]^2 \right\}, \quad (3)$$

solved iteratively with `lsqnonlin`. *Both* the real and imaginary parts are fitted, not the magnitude alone: the phase fixes the resonance location (the steep -90° crossing) and the sign of the response, and the complex residual gives two real equations per frequency, making the fit far better conditioned and unambiguous near antiresonances. The initial guesses come from simplified methods: ω_n from the magnitude peak; ζ_n from the half-power bandwidth (below); $A_{jk}^{(n)}$ from the quadrature (imaginary) peak; residuals from zero.

Half-power estimate of damping. Near resonance the dominant single-d.o.f. term gives $|H(\Omega)|^2 = |A|^2/[(\omega_n^2 - \Omega^2)^2 + (2\zeta_n \omega_n \Omega)^2]$, maximal at $\Omega \approx \omega_n$. The half-power points are where the power halves, $|H| = |H|_{\text{max}}/\sqrt{2}$. For light damping this reduces to $(\omega_n^2 - \Omega^2)^2 = (2\zeta_n \omega_n^2)^2$, so the lower/upper half-power frequencies satisfy $\omega_n^2 - \Omega_1^2 = +2\zeta_n \omega_n^2$ and $\omega_n^2 - \Omega_2^2 = -2\zeta_n \omega_n^2$; subtracting,

$$\Omega_2^2 - \Omega_1^2 = 4\zeta_n \omega_n^2 \implies \zeta_n = \frac{\Omega_2^2 - \Omega_1^2}{4\omega_n^2}. \quad (4)$$

A phase-slope estimate is used as a fallback if the half-power points fall outside the band.

Mode-shape reconstruction. For a fixed input k , $A_{jk}^{(n)} = [\phi_n(x_k)/m_n] \phi_n(x_j)$: the bracket is the same for all outputs, so the identified constants across the output stations are *proportional* to $\phi_n(x_j)$ and trace the mode shape sampled at the sensors. Only the proportions are recovered – the common factor $\phi_n(x_k)/m_n$ (and the eigenproblem’s own arbitrary scale) cannot be separated from a single input column – so the shape is known up to a constant. A common normalisation (unit maximum) removes this factor and enables the comparison of Sec. 5.

Band selection. The $\pm 5\%$ band isolates one resonance; the bands do not overlap because the natural frequencies differ by factors > 5 . If a neighbouring mode fell within $\pm 5\%$, a single-mode ($n = 1$) fit would be inadequate: one would either narrow the band or, for genuinely overlapping modes, fit them jointly with $n \geq 2$.

4.3 Results

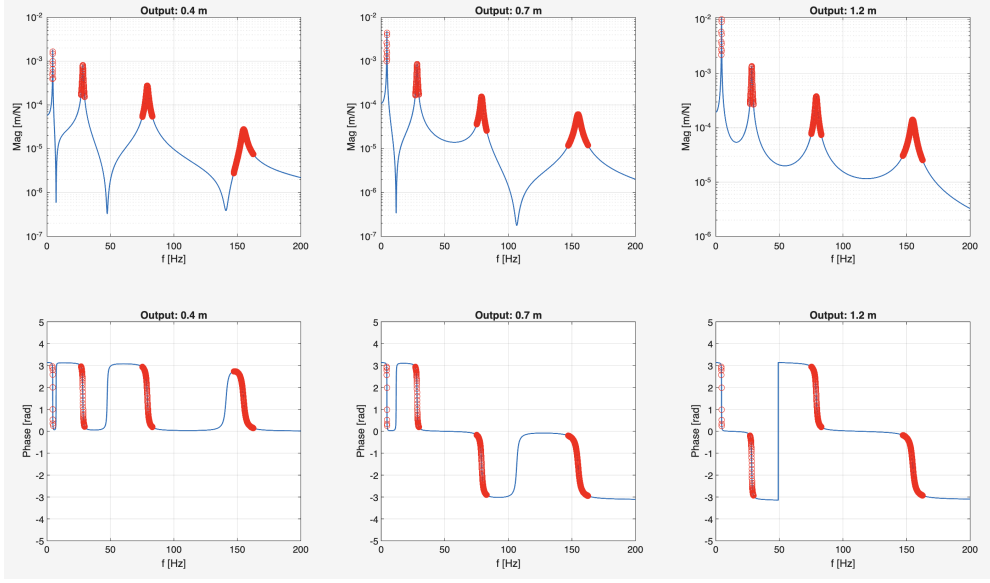


Figure 5: Identified FRFs (red markers) over the numerical ones, first configuration.

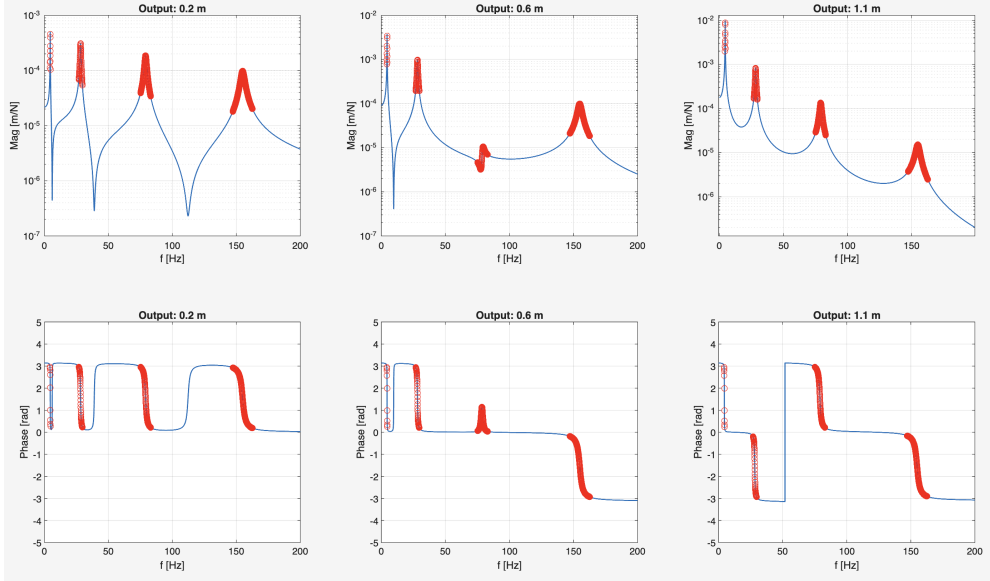


Figure 6: Identified FRFs (red markers) over the numerical ones, second configuration.

The superposition of the identified FRFs onto the numerically synthesized ones demonstrates the excellent performance of the FRF-based curve-fitting method. As visually confirmed in both configurations, the magnitude and phase of the identified curves overlap the numerical data with near-perfect accuracy across all evaluated resonances. This fit is fundamentally due to the analytical nature of the input data, which is completely free of experimental noise and leakage. Notably, the numerical optimization proves robust enough to track even the most challenging dynamics: in the second configuration, the algorithm successfully fits the severely attenuated third mode at $x_j = 0.6$ m, a condition where the signal-to-noise ratio in a real experimental setup would likely be too low to allow for a reliable identification.

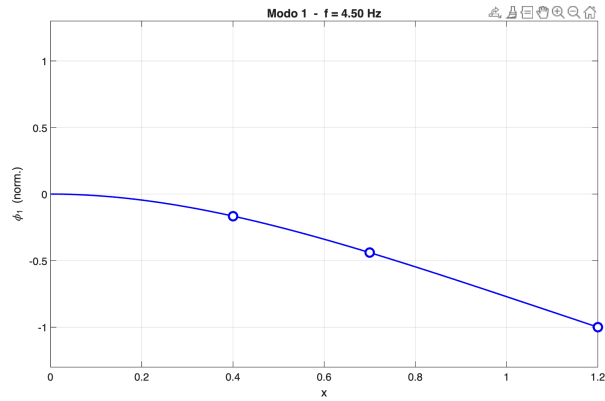
5 Comparison: simulated vs identified (first case)

Table 4: Comparison between analytical and identified modal parameters for the optimal, first configuration. The analytical frequencies are derived from the Euler-Bernoulli model, while the structural damping is uniformly set to 1%.

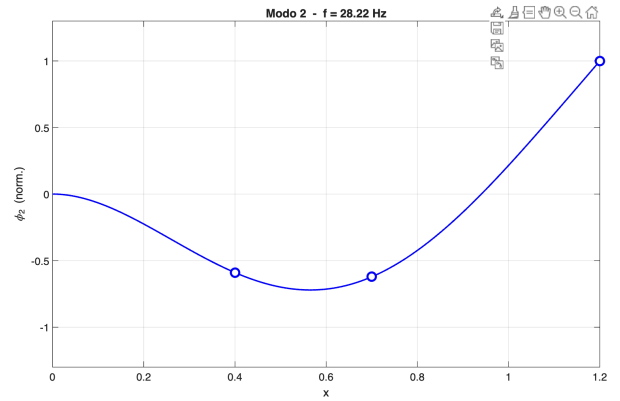
Mode	f_{model} [Hz]	f_{id} [Hz]	e_f [%]	ζ_{model} [%]	ζ_{id} [%]	e_ζ [%]
1	4.50	4.50	0.000	1.00	1.00	0.00
2	28.22	28.22	0.000	1.00	1.00	0.00
3	79.03	79.03	0.000	1.00	1.00	0.00
4	154.87	154.87	0.000	1.00	1.00	0.00

Table 5: Modal Assurance Criterion (MAC) between simulated and identified mode shapes, first configuration.

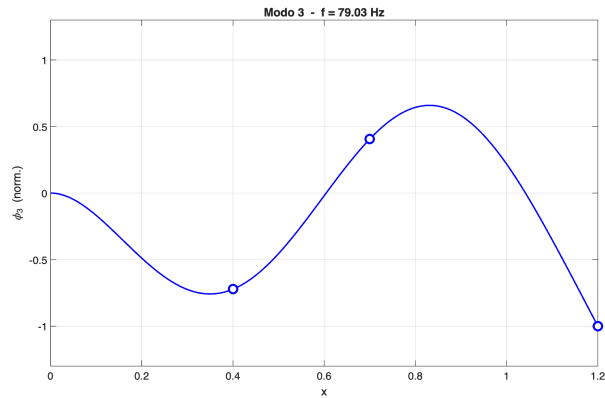
Mode	1	2	3	4
MAC	1.0000	1.0000	1.0000	1.0000



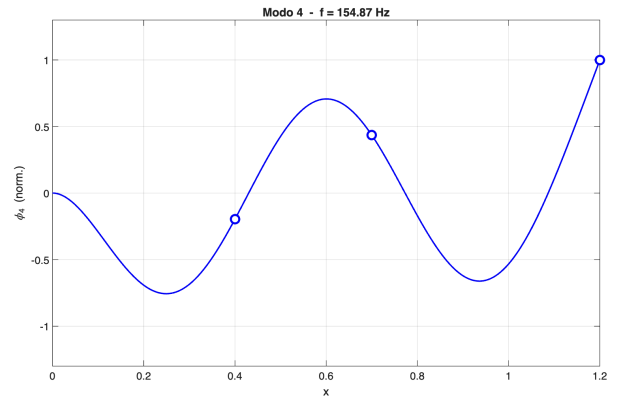
(a) Mode 1



(b) Mode 2



(c) Mode 3



(d) Mode 4

Figure 7: Simulated mode shapes (lines) vs identified points at the output stations (markers).

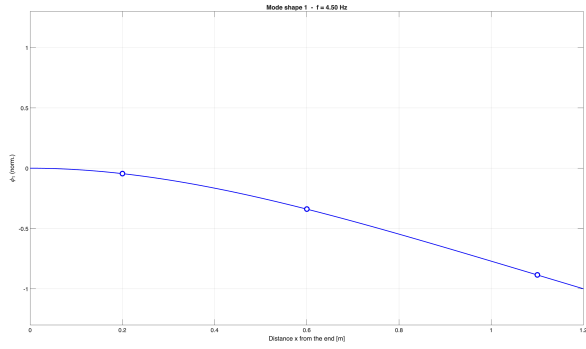
6 Comparison: simulated vs identified (second case)

Table 6: Comparison between analytical and identified modal parameters for the second configuration. The analytical frequencies are derived from the Euler-Bernoulli model, while the structural damping is uniformly set to 1%.

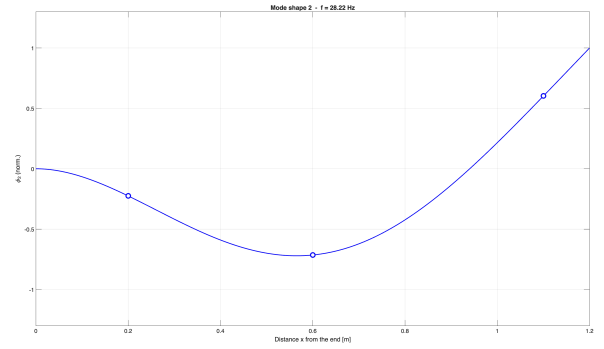
Mode	f_{model} [Hz]	f_{id} [Hz]	e_f [%]	ζ_{model} [%]	ζ_{id} [%]	e_ζ [%]
1	4.50	4.50	0.000	1.00	1.00	0.00
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4	154.87	154.87	0.000	1.00	1.00	0.00

Table 7: Modal Assurance Criterion (MAC) between simulated and identified mode shapes, second configuration.

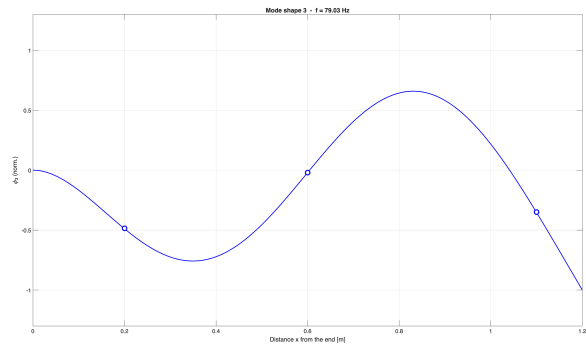
Mode	1	2	3	4
MAC	1.00	1.00	1.00	1.00



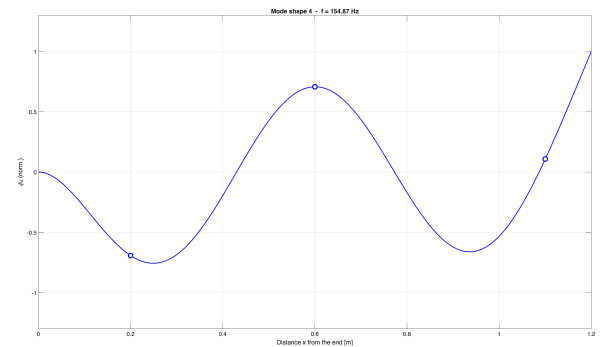
(a) Mode 1



(b) Mode 2



(c) Mode 3



(d) Mode 4

Figure 8: Simulated mode shapes (lines) vs identified points at the output stations (markers).

7 Conclusions

This report detailed the numerical synthesis and subsequent modal identification of an Euler-Bernoulli beam. A single-degree-of-freedom (SDOF) curve-fitting algorithm was employed to systematically extract the structural dynamic properties—namely natural frequencies, damping ratios, and mode shapes—from sets of synthetic Frequency Response Functions (FRFs).

The identification procedure was validated across two distinct spatial configurations. While the first setup utilized optimal sensor locations, the second configuration was deliberately designed to challenge the algorithm. By placing output sensors in close proximity to the structural nodes of specific vibration modes (e.g., at 0.6 m for the third mode), the modal participation and the prominence of the corresponding resonance peaks were drastically reduced.

Despite these inherently challenging conditions, the identification process was highly successful. The simultaneous analysis of multiple FRFs across different spatial coordinates, combined with the intrinsically ideal, noise-free nature of the analytical input data, enabled the algorithm to bypass the typical limits of low-amplitude signals. Consequently, the procedure allowed for the exact, error-free definition and reconstruction of the mode shapes at the considered output locations. This perfect geometric match is quantitatively corroborated by a Modal Assurance Criterion (MAC) consistently equal to 1.000 for all identified modes.